Metric
TT-P-2846
November 23, 1993

FEDERAL SPECIFICATION

PAINT, LATEX (RECYCLED WITH POST-CONSUMER WASTE)

The General Services Administration has authorized the use of this federal specification for all federal agencies.

1 SCOPE AND CLASSIFICATION. This specification covers latex emulsion paint containing a minimum of 50 percent post-consumer waste in the following Types, Classes and Grades:

Types	Classes	Grades
I - Interior	1 - Flat (Low sheen)	A - 40% minimum volume solids
II - Exterior	2 - Eggshell	B - 30% Minimum volume solids
III - Interior/Exterior	3 - SemiGloss	C - Utility (For graffiti abatement)

2 APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:

FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing
FED-STD-313 - Preparation and Submission of Material Safety Data Sheets (MSDS)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Code of Federal Regulations:

40CFR Part 247 - Guidelines for Procurement of Products That Contain Recycled Material

(Application for copies should be addressed to: U.S. Government Printing Office, Superintendent of Documents, Mail Stop: SSOP, Washington, DC 20402-9328)

American Society for Testing and Materials (ASTM) Standards:

D 523 - Specular gloss
D 562 - Consistency of Paints Using the Stormer Viscometer
D 1210 - Fineness of Dispersion of Pigment-Vehicle Systems
D 1296 - Odor of Volatile Solvents and Diluents
D 1308 - Effect of Household Chemicals on Clear and Pigmented Organic Finishes
D 1640 - Drying, Curing, or Film Formation of Organic Coatings at Room Temperature Materials
D 1849 - Package Stability of Paint
D 2243 - Freeze-Thaw Resistance of Water-Borne Coatings
D 2244 - Calculation of Color Differences From Instrumentally Measured Color Coordinates
D 2697 - Volume Nonvolatile Matter in Clear or Pigmented Coatings
D 3273 - Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chambe
D 3274 - Evaluating Degree of Surface Disfigurement of Paint Films by Fungal Growth or Soil and Dirt Accumulation and Adjunct No. 12-43270-00 Pictorial Standards of Coatings Defects
D 3630 - Constituents Classified as Hazardous in Protective Coatings
D 3960 - Volatile Organic Content (VOC) of Paints and Related Coatings

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Beneficial comments should be addressed to: Director, Engineering and Commodity Management Division, 9FTE-10, 400 15th St., SW, Auburn, WA 98001-6599

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3 REQUIREMENTS

3.1 **Materials.** The paint shall contain post-consumer waste as defined in 40CFR Part 247 in the quantities specified in table I. Such waste will be latex paint diverted from community waste. The manufacturer shall certify that the paint meets this requirement.

3.1.1 **Prohibited materials.** When tested as specified in table I the paint shall be free from toxic materials under normal conditions of use and shall not contain lead in excess of 0.06 percent by weight of nonvolatile, hexavalent chromates, benzene, halogenated solvents, or mercury in excess of 10 parts per million for Types I and III or 50 parts per million for Type II.

3.2.1 **Condition in container.** When examined as specified in table I, the paint shall show no evidence of putrefaction, hard settled pigment, or corrosion of the container. The paint shall be dispersible to a uniform condition by not more than 5 minutes hand stirring without decanting and remixing.

3.2.2 **Color.** When tested as specified in table I at complete hiding, the Grade A and B paint shall match the color specified within a color tolerance, delta E, of 2.5. The color of the Grade C paint shall be as agreed upon by buyer and seller.

3.2.3 **Accelerated storage.** After storage at 52°C (125°F) for 30 days as specified in table I, a sealed, filled one-liter (one-quart) can of paint shall show no coagulation or hard settled pigment. The paint shall be dispersible to a uniform condition and shall pass the application properties tests specified in 3.2.5.

3.2.4 **Freeze-thaw stability.** When tested as specified in table I for 3 freeze-thaw cycles, the paint shall show no coagulation or flocculation, the consistency shall not change more than 8 KU, and the paint shall pass the brushing properties test in 3.2.5.

3.2.5 **Application properties.** When tested as in 4.3.1, the paint shall brush, roll, and spray easily; and shall dry to a smooth uniform film free from lap marks, excessive brush marks, orange peel, craters or dusting.

3.2.6 **Odor.** When tested as specified in table I, the odor of the paint in the can and during application shall not be irritating. The paint shall have no residual odor after 48 hours air drying.

3.2.7 **Alkali resistance.** When tested as specified in 4.3.3, the paint shall show no blistering or re-emulsification immediately after test. After 24 hours recovery, the film shall show no change in hue or hardness when compared with the untested portion of the paint film.

3.2.8 **Flexibility.** When tested as specified in 4.3.2, the paint film shall not crack or flake.

3.2.9 **Scrub resistance.** The film shall not be worn through to the panel in fewer than 300 cycles for grade A or 150 cycles for grade B when tested as specified in table I.

3.2.10 **Biological growth.** When tested as specified in table I, the paint shall attain a surface disfigurement rating of 8 or greater when evaluated against Adjunct No. 12-432740-00 specified in ASTM D 3274. A certificate of compliance may be accepted from the supplier that a biocide has been used in concentrations which has been tested in similar formulations and has passed this requirement.

3.3 **Quantitative requirements.** The paint shall meet the requirements specified in table I for the type, class and grade specified.

TABLE I. Quantitative requirements and test references

Characteristic	Numeric Limits	Requirement Paragraph	Test Paragraph	Test Method[1]
Requirements for all Grades				
Materials	—	3.1	—	—
Post-consumer waste percent	—			
Grades A and B	50 min			
Grade C	90 min			
Prohibited materials	—	3.1.1	—	D 3630
Condition in container	—	3.2.1	—	3011
Color	—	3.2.2	—	D 2244
Accelerated storage	—	3.2.3	—	D 1849

1 Test methods prefixed with a letter are ASTM methods. Others are FED-STD-141 test methods.

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TABLE I. Quantitative requirements and test references (Continued)

Characteristic	Numeric Limits	Requirement Paragraph	Test Paragraph	Test Method[1]
Requirements for all Grades				
Freeze-thaw stability	—	3.2.4	—	D 2243
Application properties	—	3.2.5	4.3.1	2112, 2131, 2141
Odor	—	3.2.6	—	D 1296
Dry through, hr	—	—	—	D 1640
Classes 1 and 2	2 max			
Class 3 —	8 max			
Consistency, KU	80 - 100	—	—	D 562
Volatile organic compound content (less water and exempt solvents) g/L (lb/gal)	200 max (1.67)	—	—	D 3960
Contrast ratio at 9.8 m^2/L (400 ft^2/gal)	—	—	—	4121[2]
Reflectivity 80 and above	0.92 min			
76 - 79	0.93 min			
72 - 75	0.94 min			
68 - 71	0.95 min			
61 - 67	0.96 min			
60 and lower	0.98 min			
Requirements for grades A and B				
Alkali resistance	—	3.2.7	4.3.2	D 1308
Flexibility, inch	1/4	3.2.8	4.3.3	6221
Scrub resistance	—	3.2.9	—	D 2486
Grade A	300 min			
Grade B	150 min			
Biological growth	8 min	3.2.10	—	D 3273, D 3274
Total solids, % volume of paint		—	—	D 2697
Grade A	40 min			
Grade B	30 min			
Fineness of dispersion		—	—	D 1210
Class 1	3 min			
Classes 2 and 3	4 min			
Gloss			4.3.4	D 523
85° gloss, Class 1	10 max			
60° gloss, Class 2	15 - 25			
60° gloss, Class 3	30 - 50			

1 Test methods prefixed with a letter are ASTM methods. Others are FED-STD-141 test methods.
2 Procedure B, method B

3.4 **Special marking.** Each container shall be marked with the percentage of post-consumer recycled waste. Each container and shipping container shall be marked:

"DO NOT FREEZE"

3.5 **Material Safety Data Sheet.** A Material Safety Data Sheet shall be prepared in accordance with FED-STD-313.

4 QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or order, the contractor is responsible for the performance of all inspection requirements specified herein using facilities approved by the Government. The Government reserves the right to perform any of the inspections set forth herein when deemed necessary to assure that the paint conforms to prescribed requirements.

4.2 **Classification of inspections.** Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.2.1).

4.2.1 **Preparation for delivery.** A random sample of filled containers shall be selected in accordance with MIL-STD-105, inspection level S-2, acceptable quality level (AQL) 2.5 percent defective, and examined for compliance with 3.4 and section 5.

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4.3 **Quality conformance inspection.** The paint shall be tested in accordance with the methods specified in Table I and as otherwise specified herein to determine compliance with the requirements of section 3. Unless otherwise specified, all tests shall be conducted at conditions specified in section 9 of FED-STD-141. Failure of any test shall be cause for rejection of the lot from which the sample was taken.

4.3.1 **Application properties.**

4.3.1.1 **Brushing properties.** Brush the paint in accordance with method 2141, FED-STD-141 at a spreading rate of 9.8 m^2/L (400 ft^2/Gal) on a primed gypsum wallboard panel prepared in accordance with method 2081, FED-STD-141. Evaluate during brushing and after drying for compliance with 3.2.4.

4.3.1.2 **Roller coating properties.** Roll the paint in accordance with method 2112, FED-STD-141 at a spreading rate of 9.8 m^2/L (400 ft^2/Gal). Evaluate during rolling and after drying for compliance with 3.2.4.

4.3.1.3 **Spraying properties.** Spray the paint in accordance with method 2131, FED-STD-141. Evaluate during spraying and after drying for compliance with 3.2.4.

4.3.2 **Alkali resistance.** Prepare and dry a panel as specified in ASTM D 2486. Test the dried film using 0.1 N trisodium phosphate solution in accordance with paragraph 6.2, ASTM D 1308. Wash the reagent off with distilled water after 4 hours contact time and examine the film immediately and after 24 hours recovery.

4.3.3 **Flexibility.** Prepare the test panel in accordance with method 2012, FED-STD-141. Supplement the panel cleaning with an additional cleaning with abrasive soap so that the surface shows no water break. Draw down the paint on the clean, dry panel with a film applicator to obtain a dry film thickness of 25 +/- 2 um (0.001 +/- 0.0001 inch). Air dry 18 hours, bake at 105 +/- 2°C (221 +/- 4°F) for 3 hours, and cool 1/2 hour at room temperature. Bend over a 6.35 mm (1/4 inch) diameter cylindrical mandrel and examine under a magnification of 7 diameters in accordance with method 6221, FED-STD-141.

4.3.4 **Gloss.** Draw down the paint using a film applicator which will produce a wet film thickness of 75 +/- 2 um (0.003 +/- 0.0001 inch) on a plane glass panel. Determine 85° or 60° specular gloss in accordance with ASTM D 523 after 48 hours drying in a dust-free environment.

5 PREPARATION FOR DELIVERY

5.1 **Packaging, packing, and marking.** The paint shall be furnished in quantities specified (see 6.2). The packaging, packing, and marking shall be as specified (see 6.2).

6 NOTES

6.1 **Intended use.** This emulsion paint is intended is for use on interior or exterior wallboard, concrete, stucco, masonry, and wood. Chalk and loose paint should be removed before painting. New wood should be primed with a suitable primer. Application temperatures should be above 10°C (50°F) to insure proper drying and film formation.

6.2 **Ordering data.** Purchasers should include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, Class, Grade and color required (see 1).
(c) Packaging, packing, and marking required.
(d) Size of container and quantity required.

6.3 **Bid evaluation.** When specified by the contracting officer, competing offers will be evaluated using the following formula with the lowest evaluated offer being considered as low bid:

Evaluated offer = offer price/AxB where:

A = volume solids percentage divided by 100
B = post consumer waste percentage divided by 10

Example: offer price $10.00 per gallon, 50% volume solids, 50% post-consumer waste

Evaluated offer = $10.00/(50/100)x(50/10) = $4.00 per gallon

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